

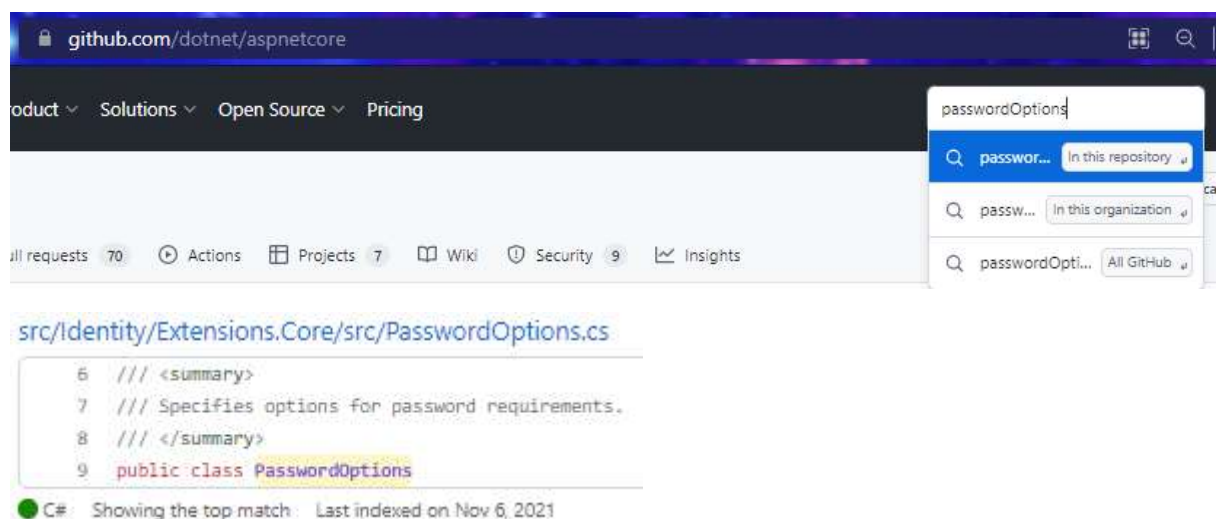
68 - Identity password complexity

Default Settings in PasswordOptions

```
public class PasswordOptions
{
    public int RequiredLength { get; set; } = 6;
    public int RequiredUniqueChars { get; set; } = 1;
    public bool RequireNonAlphanumeric { get; set; } = true;
    public bool RequireLowercase { get; set; } = true;
    public bool RequireUppercase { get; set; } = true;
    public bool RequireDigit { get; set; } = true;
}
```

<https://github.com/aspnet/aspnetcore>

<https://github.com/dotnet/aspnetcore> - Open Source:



The screenshot shows the GitHub repository page for `dotnet/aspnetcore`. The search bar at the top right contains the text `passwordOptions`. Below the search bar, there are three search results: `password...` (In this repository), `passw...` (In this organization), and `passwordOpti...` (All GitHub). The file `src/Identity/Extensions.Core/src/PasswordOptions.cs` is highlighted. The code snippet shows the `PasswordOptions` class with the following properties:

```
6 /// <summary>
7 /// Specifies options for password requirements.
8 /// </summary>
9 public class PasswordOptions
```

At the bottom, it says "C# Showing the top match: Last indexed on Nov 6, 2021".

```

4 namespace Microsoft.AspNetCore.Identity;
5
6 /// <summary>
7 /// Specifies options for password requirements.
8 /// </summary>
9 public class PasswordOptions
10 {
11     /// <summary>
12     /// Gets or sets the minimum length a password must be. Defaults to 6.
13     /// </summary>
14     public int RequiredLength { get; set; } = 6;
15
16     /// <summary>
17     /// Gets or sets the minimum number of unique characters which a password must contain. Defaults to 1.
18     /// </summary>
19     public int RequiredUniqueChars { get; set; } = 1;
20
21     /// <summary>
22     /// Gets or sets a flag indicating if passwords must contain a non-alphanumeric character. Defaults to true.
23     /// </summary>
24     /// <value>True if passwords must contain a non-alphanumeric character; otherwise false.</value>
25     public bool RequireNonAlphanumeric { get; set; } = true;

```

Може да ги персонализираме в Startup -> ConfigureServices

```

public void ConfigureServices(IServiceCollection services)
{
    services.AddDbContextPool<AppDbContext>(
        options => options.UseSqlServer(_config.GetConnectionString(

    services.AddIdentity<IdentityUser, IdentityRole>()
        .AddEntityFrameworkStores<AppDbContext>());

    services.Configure<IdentityOptions>

    services.AddMvc(option => option.EnableEndpointRouting = false)

```

Go to Definition:

```

namespace Microsoft.AspNetCore.Identity
{
    ...public class IdentityOptions
    {
        public IdentityOptions();

        ...public ClaimsIdentityOptions ClaimsIdentity { get; set; }
        ...public UserOptions User { get; set; }
        ...public PasswordOptions Password { get; set; }
        ...public LockoutOptions Lockout { get; set; }
        ...public SignInOptions SignIn { get; set; }
        ...public TokenOptions Tokens { get; set; }
        ...public StoreOptions Stores { get; set; }
    }

```

Можем да персонализираме изискванията по един от двата начина:

Override Password Default Settings

```
services.Configure<IdentityOptions>(options =>
{
    options.Password.RequiredLength = 10;
    options.Password.RequiredUniqueChars = 3;
    options.Password.RequireNonAlphanumeric = false;
});
```

OR

```
services.AddIdentity<IdentityUser, IdentityRole>(options =>
{
    options.Password.RequiredLength = 10;
    options.Password.RequiredUniqueChars = 3;
    options.Password.RequireNonAlphanumeric = false;
})
.AddEntityFrameworkStores<AppDbContext>();
```

```
services.Configure<IdentityOptions>(options =>
{
    options.Password.RequiredLength = 10;
    options.Password.RequiredUniqueChars = 3;
});
```

```
public void ConfigureServices(IServiceCollection services)
{
    services.AddDbContextPool<AppDbContext>(
        options => options.UseSqlServer(_config.GetConnectionString(

    services.AddIdentity<IdentityUser, IdentityRole>()
        .AddEntityFrameworkStores<AppDbContext>();

    services.Configure<IdentityOptions>(options =>
    {
        options.Password.RequiredLength = 10;
        options.Password.RequiredUniqueChars = 3;
    });

    services.AddMvc(option => option.EnableEndpointRouting = false)
```

Има възможност да направим същото директно с AddIdentity:

```

public void ConfigureServices(IServiceCollection services)
{
    services.AddDbContextPool<AppDbContext>(
        options => options.UseSqlServer(_config.GetConnectionString('

services.AddIdentity<IdentityUser, IdentityRole>(options =>
{
    options.Password.RequiredLength = 10;
    options.Password.RequiredUniqueChars = 3;
}).AddEntityFrameworkStores<AppDbContext>());

services.AddMvc(option => option.EnableEndpointRouting = false)

```

```

services.AddIdentity<IdentityUser, IdentityRole>(options =>
{
    options.Password.RequiredLength = 10;
    options.Password.RequiredUniqueChars = 3;
})

```

Когато се опитаме да регистрираме нов потребител, вече имаме обновени изискванията:

User Registration

- Passwords must be at least 10 characters.
- Passwords must have at least one digit ('0'-'9').
- Passwords must have at least one lowercase ('a'-'z').
- Passwords must have at least one uppercase ('A'-'Z').
- Passwords must use at least 3 different characters.